

FALCON's Response to SAB Express RFQ for the supply of SWEDI Sorter of Throughput 4000 PPH

May 9, 2023



www.falconautotech.com

Kind Attention –

Mr. Turab Rahaman

SAB Express

Offer Ref: F22-00317; Date 09.05.2023

Subject – Revised Proposal for the supply of SWEDI sorter of throughput 4000 PPH.

Dear Sir,

We are pleased to submit our Revised Offer for Supply of SWEDI Sorter

As you will note, we have studied your requirements in great depth and, along with the information collected during the workshops and calls with you, we have put together a detailed technical proposal.

In subsequent sections, we have highlighted the capabilities and experiences of Falcon Autotech as well as the capabilities of the proposed software.

To conclude, I would like to add my personal commitment, on behalf of Falcon Autotech. As we move through the RFP process, please do not hesitate to contact me and my team. We will be pleased to assist you for any further information or clarifications that you might have.

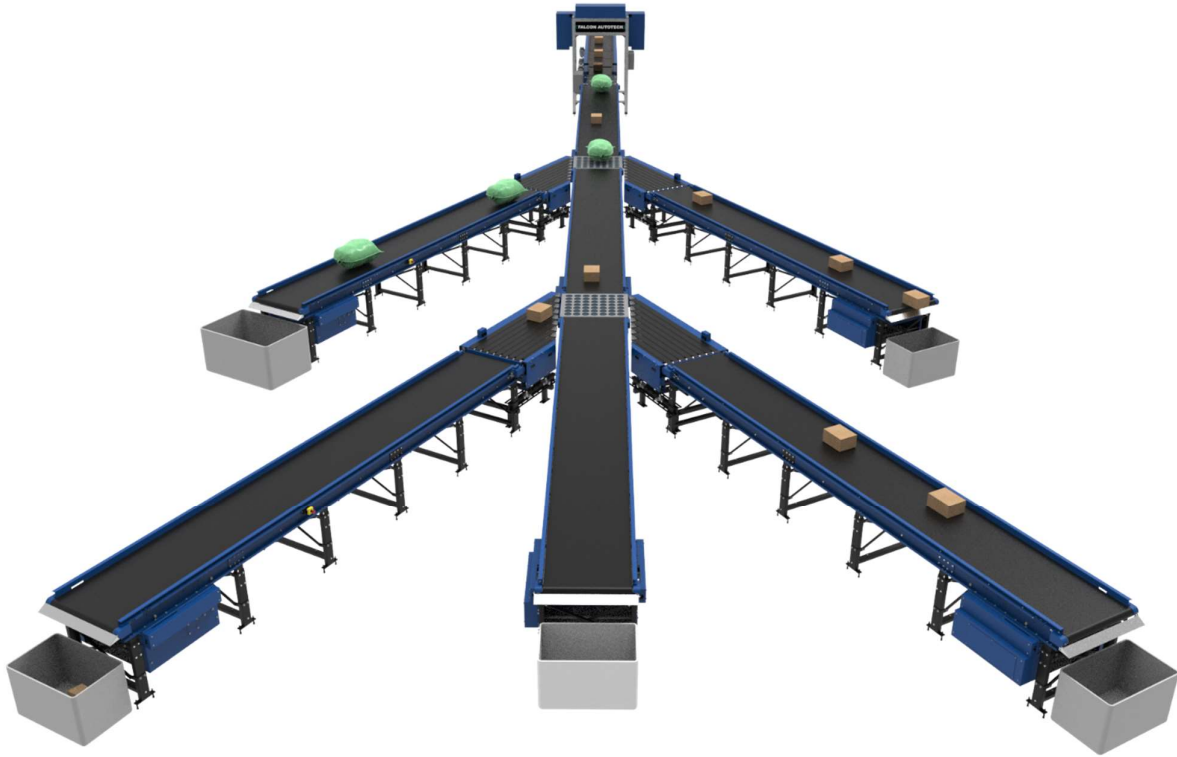
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Website : www.falconautotech.com
Linked ID : <https://www.linkedin.com/company/falconautotech/mycompany/>

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Response Quotation for Swivel Wheel Diverts



Proposal Reference - F22-00317.01

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India.

Contact – Sanyog Singh
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1. Glossary

S. No.	Term	Description
1	RFQ	Request For Quote
2	PPH	Parcels Per Hour
3	ICR	Intelligent Character Recognition
4	PLC	Programmable Logic Controller
5	IT	Information Technology
6	BOQ	Bill Of Quantity
7	I/O	Input/ Output
8	PC	Personal Computer
9	UPS	Uninterrupted Power Supply

2. Company Profile

THE COMPANY

OUR VISION

Create a Global impact on the society by facilitating World's leading Organizations run their Supply Chains efficiently by leveraging technology enabled products and Solutions and, in the process, establish Falcon Autotech as one of the World's most respected and trusted brand in Intra-logistics Warehouse Automation Space.

ABOUT US

The team started out in 2004 solving Special Purpose Automation problems for clients and later pivoted to building Standard Technology Stack spanning across hardware, firmware and Software to tackle bigger Supply Chain Problems.

Over the last few years, Organization has made rapid strides and has carved out a niche in some of the world's most cutting edge technologies: Sortation, Robotics, Conveying, Vision Systems and IOT.

TEAM

Organizations are built and defined by the people they are able to attract, cultivate and retain and the talent management philosophy at Falcon Autotech could not agree more.

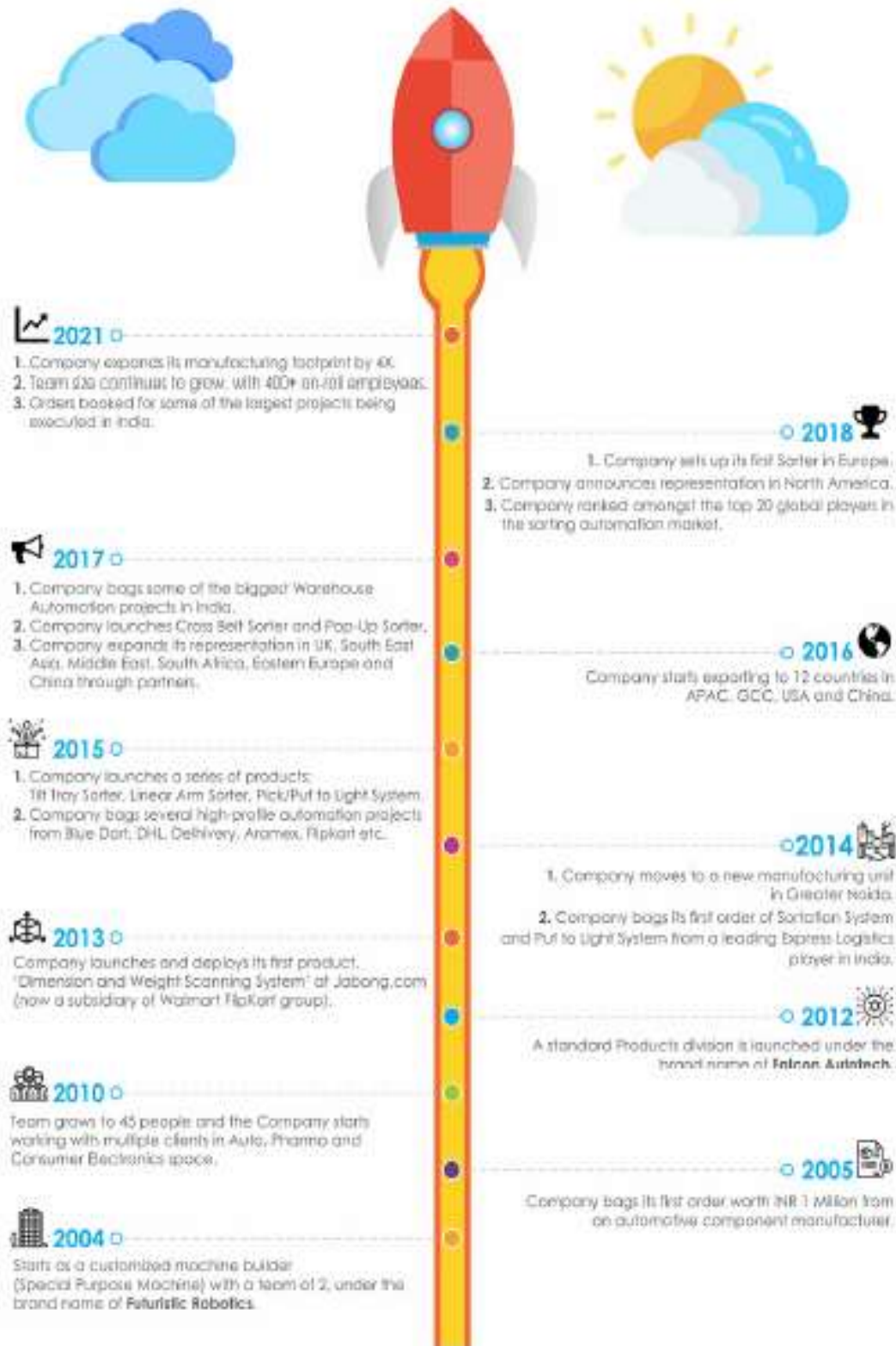
At, Falcon Autotech we take a lot of pride in being the preferred career choice for some of the finest talent in the country across multiple domains such as:

- ❖ Supply Chain Solution Designing and Consultation
- ❖ Hardware Design
- ❖ Electrical Design
- ❖ Electronics Design
- ❖ Software Development
- ❖ Manufacturing

Giving us an unprecedented ability harness true potential generated by culmination of Supply Chain Know How, Hardware and Software under one roof.



COMPANY HISTORY



KEY STATISTICS

**04**Product Lines

**525+**On-Roll Employees

**290+**Engineering Team

**400000+ Sq Ft**Manufacturing Space

**20+**Tier A Clients

**15+**Countries with Live Installations

**1700+**Total Installed Systems Globally

**35**Representations in 4 Continents and 35 Countries

**1 Million+**Sorts Per Hour of combined Sorting Capacity
installed Globally

OUR PRODUCT PORTFOLIO



SORTATION SOLUTIONS

01

- ❖ Cross Belt Sorter
- ❖ Tilt Tray Sorter
- ❖ Sliding Shoe Sorter
- ❖ Pop Up Sorter
- ❖ Swivel Wheel Sorter
- ❖ Arm Sorter
- ❖ Pusher Sorter
- ❖ Drop Down Conveyor Sorter

PICK/PUT TO LIGHT SYSTEMS

02

- ❖ Packet Sortation CEP
- ❖ Multi-Line Order Consolidation for E-Commerce
- ❖ Order Picking for Food, FMCG, Pharmaceutical
- ❖ Store Order Fulfillment for Fashion & Grocery
- ❖ Spare Part Distribution for Automobile, Electronics
- ❖ Kitting for Automotive, Consumer Electronics, Heavy Machinery Companies
- ❖ Multi Order Picking Cart



DIMENSION & WEIGHT SCANNING SYSTEMS

03

- ❖ Static Dimension & Weight Scanning Systems
 - Cubizon-R
 - Cubizon-R ECO
 - Cubizon-R THRU
 - Cubizon-R Cross
- ❖ Dynamic Conveyor Based Systems
 - Semi-Automatic Dynamic Profilers
 - Fully-Automatic Dynamic Profilers

OUR PRODUCT PORTFOLIO



CONVEYOR AUTOMATION SOLUTIONS

04

- ❖ Belt Conveyors
- ❖ Roller Conveyor
- ❖ Modular Belt Conveyors
- ❖ Inclines/ Decline Conveyors
- ❖ Conveyor Bends
- ❖ Traffic Control Merges
- ❖ Product Accumulation
- ❖ Code Scanning
- ❖ Automated Label Print And Apply
- ❖ Spirals
- ❖ In-line Weighing And Dimensioning
- ❖ In-line Carton Packaging
- ❖ In-Line Case Packaging Systems (Taping, Strapping, Shrink Wrapping)
- ❖ In Line Pouch On Demand Packaging (Cold Seal Packing)

SOFTWARE SUITE

05

❖ ControlIT - Warehouse Automation Control System

- Sorter Control Package
- DWS Control Package
- PPTL Control Package
- Dashboard Package

❖ AMS (Automation management System)

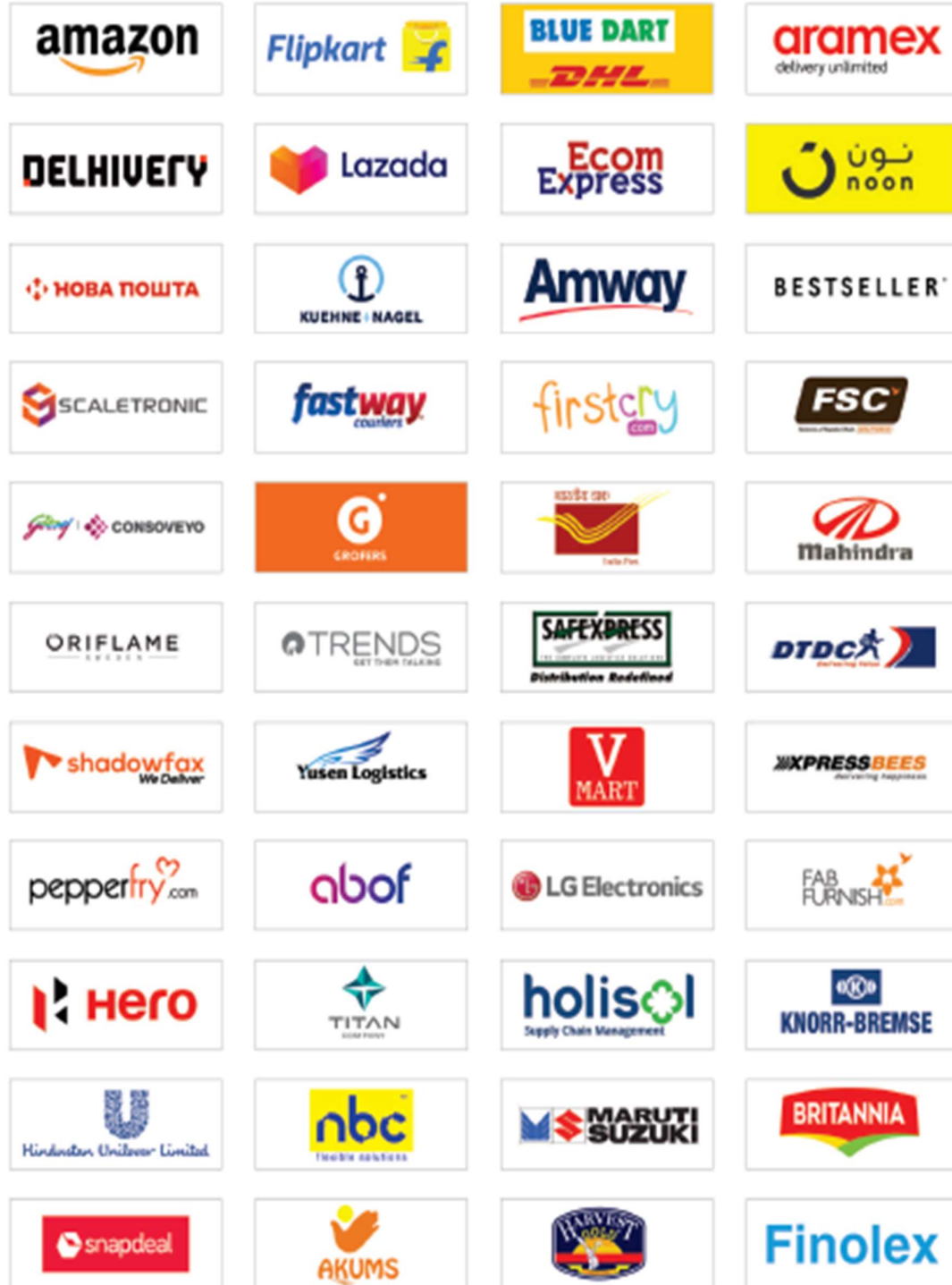
Falconautotech's unique and proprietary Automation Systems Management Engine: **AMS** is a one of its kind software platform available in the market today that opens up tremendous avenues for:

- Predicting and avoiding unplanned down times.
- Increased visibility and control into health, maintenance and functioning of the equipment.
- Efficient Spares Management System.
- Seamless Knowledge transfer for system maintenance and troubleshooting.



SOME OF OUR GLOBAL CLIENTS

We believe in delivering high quality and standard worldwide. Here is a gist of our esteemed clients across various industry segments.



INDUSTRY SOLUTIONS

As a leading player in Intra Logistics Automation Space, Falcon Autotech continuously strives to improve the operational efficiency and accuracy for its clients through its domain knowledge consultancy and experience in addition to its wide range of Products and solutions.

In order to be able to live up to the high expectations set forth by our clients, the team at Falcon Autotech realizes the importance of taking up selective applications in Select Industries that lie in the Forte of Falcon Autotech and deliver world class projects in return.

FOCUS INDUSTRIES



E-COMMERCE RETAIL



COURIER EXPRESS POSTAL



FASHION DISTRIBUTION



FFP (FOOD, FMCG, PHARMA) DISTRIBUTION



SPARE PARTS DISTRIBUTION



GROCERY DISTRIBUTION

FOCUS APPLICATIONS FOR FALCON AUTOTECH

- ❖ Piece Picking and Handling
- ❖ Case Picking and Handling
- ❖ Parcel Handling

CLIENT TESTIMONIALS

“ Working with Falcon Autotech has been a very delightful experience for us, BLUE DART being a leader in the express logistics business, we expect the same kind of responsiveness, reliability, commitment & passion from our partners and Falcon has always delivered to their promise on all the solutions such as the Auto Sorter, Weight Dimension Labelling machine, Parcel Locker etc. All the technology interventions have been business critical for us.



Anil Khanna, Managing Director
DHL BLUE DART

“ Falcon is an organisation that has shown great agility and has helped us improve our service capabilities. Their sortation systems have helped us deal effectively with our growing volumes. We have found the Falcon team quite responsive to feedback and quick to incorporate changes into their systems that have helped build better, customized solutions for Delhivery.



Sahil Barua, CEO
DELHIVERY

“ Falcon Autotech has been efficient and proactive, particularly in meeting deadlines and their technical advancements. Falcon Autotech strike just the right balance between being a flexible, easy to deal with firm yet having all the key strengths and attributes. Every member of the team is honest, reliable, responsive, personable and very knowledgeable. I wish them good luck for all the future endeavours.



Sanjeev Saxena, Director
ECOM EXPRESS

“ I believe you guys earned it with your hard work and dedication. I believe the products you have designed demonstrate innovation and lateral thinking. Also the management has a flexible and the customer centric approach. It has been a delight to work with you and the team.



Amitava Saha
CEO, firstcry.com

3. Falcon's Experience and Achievements in Sortation Space Globally

- Ranked among **Top 20 Sortation System Suppliers** globally. Only company from APAC/EMEA Region.
- Currently possess one of the **World's largest portfolio in Sortation Technologies**: 7 In-house technologies.
- Total installed capacity of **6 Million Parcels per day** worldwide.
- Only company to be able to offer a **Fully Integrated AMS**.



Incas-SSI Schaefer merger creates leading player in Italy's intralogistics market

TOP STORIES

CROWDFUNDING

BUSINESS

FINANCE

CRYPTO

COMMODITIES

LUXURY

TOUCHPOINTS

SO

Although, the transportation management segment of the logistics automation industry is expected to have a higher CAGR. This is due to factors associated with the growing need to deliver goods in a timely manner and to reduce transportation costs.

Moreover, automotive will be the leading segment of the industry from 2018 to 2023, with an increasing demand for automated production systems and flexible logistics systems used in manufacturing and supply of vehicle fleets.

The MarkestandMarkets research report has identified SSI Schaefer as one of the key players in the logistics automation market worldwide during the forecast period, along with Dematic Corporation (US), Murata Machinery, Ltd. (Japan), Jungheinrich AG (Germany), and Falcon Autotech (India), among others.

This year's largest market for logistics automation is estimated to be North America, according to the research report. Regardless, Incas and SSI Schaefer's merger could help them compete in their main market.

(Featured image by [DepositPhotos](#))

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Not secure | worldindustryinsights.com/30269/global-parcel-sortation-system-market-analysis-size-share-growth-trends-and-forecast-2019-2026-siemens-vanderlande-beumer-group



Global Parcel Sortation System Market Analysis, Size, Share, Growth, Trends, and Forecast 2019-2026: Siemens, Vanderlande, Beumer Group, Intelligrated, Bastian Solutions, Fives

BY ANN.CASTRO ON AUGUST 12, 2019

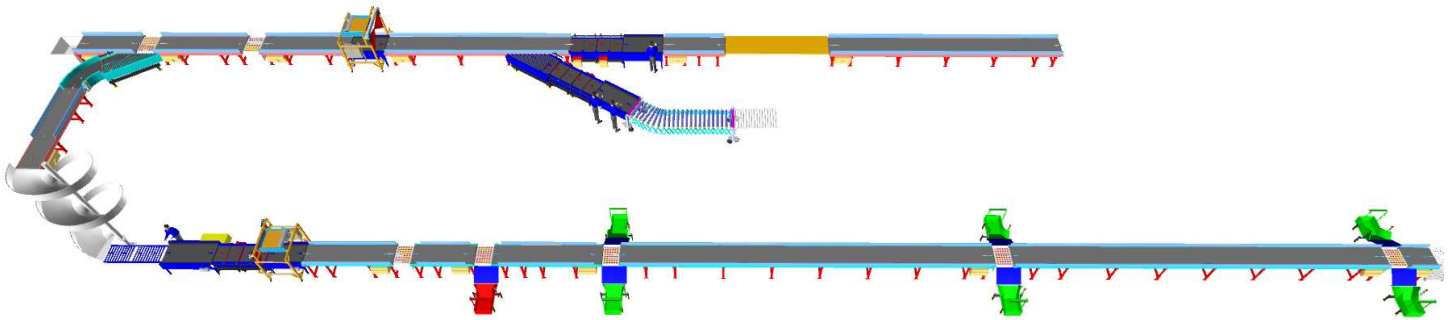
The global "[Parcel Sortation System market](#)" research report is a pervasive research report that touches the most vital parts of the Parcel Sortation System market that is necessary to be grasped by a professional or even a layman. The market research report enlightens one regarding few of the important aspects such as an overview of the Parcel Sortation System product, the growth factors enhancing or hampering its development, application in the various fields, major dominating companies, genuine facts, economic situation, and geographical analysis. The research report endows the data concerning the dynamics that propel the growth as well as the supply and demand chain of the product on a global basis.



Click here to access the report: www.reportsbuzz.com/request-for-sample.html?repid=50822

The international and global stand of the Parcel Sortation System market is also briefly mentioned in the research report based on the performed statistical and thorough market analysis. The information mentioned in the research report gives a qualitative and quantitative view of the overall market. The statistical analysis of the market helps analyze the supply, demand, production, maintenance, and storage costs of the product. The data regarding some of the prevalent players Siemens, Vanderlande, Beumer Group, Intelligrated, Bastian Solutions, Fives, Dematic, Interroll, Muratec, Invata Intralogistics, Viastore, Okura Yusoki, GBI Intralogistics, Equinox Mhe, Pitney Bowes, Falcon Autotech, OCM SRL, Solystic, Bowe Systec, Intralox are also detailed in the current case study.

4. Layout View



Process notes

1. Swivel wheel diverter is to be installed at ground floor and the basement.
2. The system has 2 infeed points.
3. At First Infeed Point:
 - a) Shipments are loaded on to the infeed conveyor in a singulated manner.
 - b) It will pass through the X-ray machine.
 - c) The operator will check the shipments, reject the wrong ones, and orientate the rest in a manner that the bar code is on the top.
4. At Second Infeed Point:
 - a) The pallets are fed into the system.
 - b) It will pass through the X-ray machine.
 - c) The operator will bring down the pallet from the conveyor, With the help of pallet handler. break it and fed it to the sorter in a singulater manner ensuring barcode facing top.
 - d) If the X-ray machine of first infeed will not work, then the shipments will also come from the second (pallet) x-ray machine. These shipments are being transferred from x-ray machine to the feedline through powered flexible roller conveyor.
 - e) Also, whenever the throughput number high pallet x- ray feedline used for shipment feeding.
3. These shipments are then scanned by automatic scanner.
5. The first sorting point is at ground floor. Rest of the shipments will be moved to the basement with the help of spiral chute.
6. In the basement, there will be an operator who will orientate the shipment and ensure that the barcode is at the top.
8. Then these shipments then further scanned by automatic barcode scanner.
9. Based on 5 locations, the shipments will be sorted.
10. There is rejection chute as well for rejected shipments.

5. Handled Parcel Spectrum

a. Parcel size loadable on the sorter

Product Range					
Transported Items	Size	Length (mm)	Width (mm)	Height (mm)	Weight (Kg)
Carton boxes, Poly Bags	Maximum	900	500	500	35
	Minimum	150	150	50	0.5

b. Parcels to be loaded on Sorter shall have the following characteristics:

1. Centre of Gravity of item must not move during conveyance or sorting.
2. Item must not have magnetic content, otherwise behavior of parcel cannot be guaranteed.
3. Liquid or fragile material, to avoid breaking, spillage or leakage, such as wine bottles, metal cans of paint are designated as non- conveyable items.
4. Parcels shall be perfectly and safely packaged: protrusion or open surfaces are not allowed.
5. Plastic ropes shall be perfectly adherent to the surface of the package.
6. Loose parts on loads and load carriers, such as adhesive tape, stickers, slips of paper, straps, wrap foil etc. are designated as non- conveyable items.
7. All items with the risk of being damaged during the transport on an automatic sorting system or damaging the sorting system. They must be robust enough to avoid disintegration of container material and loose of contents in the sorting process.
8. Item packaging shall have enough grip to be handled on the belts during the acceleration and referencing phases.
9. Items shall not have slippery surfaces and must be able to withstand acceleration of the items on the belt during the start-stop phases (accelerations up to 0.5 g shall be assured without any sliding or tumbling of the items on the belt conveyor).
10. The parcels must have at least one flat and regular surface providing enough stability during conveyance.
11. Unstable items with a risk to roll or tumble on the sorting system, such as spherical or cylindrical items
12. All shapes are permitted except spherical, cylindrical, or alike unstable items & shapes.
13. All usual packaging materials are permitted (including paper, carton, plastics, plastic foil, rope, tape, textile, and wood)

c. Parcel not loadable on the sorter

All products that are not within the range as described here, are considered non- conveyable products, and must be taken out of the main sorter flow by the operators.

1. Items that have a spherical or cylindrical shape.
2. Items that are packed in material that can damage the conveyors or the sorter.
3. Items that have sharp points (e.g. Nails) or sharp edges, that can damage the conveyors or the sorter.
4. Fragile parcels with contents not sufficiently secured
5. Items that have been classified as dangerous are designated.
6. Wet items are designated.
7. Items with anti-slip treatment.
8. Items with protruding parts.
9. Items with sharp edges.
10. Inadequately packed items that could be damaged during automatic transportation.
11. Electrostatically loaded items.

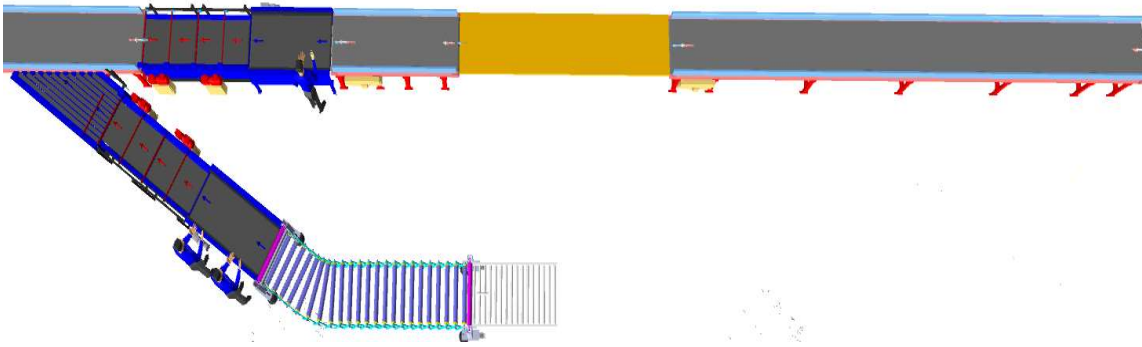
d. Pictures of Sortable Shipments

PARCEL TYPE	DESCRIPTION
	Strongly Strapped A parcel banded with strapping material.
	Pre Manufactured Plastic Courier Pack <A3 in size A parcel shrouded in a plastic coating and is a manufactured product of a pre-determined size.
	Pre Manufactured Plastic Courier Pack > A3 in size A parcel shrouded in a plastic coating and is a manufactured product of a pre-determined size. Loose part with thickness under to minimum allowed (10 m) may create problems and are not recommended to be processed.
	Other Plastic Covered Item A cardboard parcel coated in a plastic material, for example bin liner, shrink-wrap and stretch-wrap.
	Paper Covered Item A cardboard parcel coated in a packing paper.
	Cloth Covered Item Often Import parcels that are of irregular shape.
	Wine Box A robust, cardboard parcel that is designed to protect its fragile contents from breakage and has a high weight to size ratio.
	Cardboard Box A cardboard parcel is often the norm.
	Totes A large plastic, lidded box.
	Flats A parcel that often contains documents.
	Poly Bags It is a plasticized weave bag, used as a containerization solution, holding a number of other parcels within it.

6. Key Modules

6.1 Automatic Feedlines

Fully automatic feedline is used in the system to induct parcels directly on to the sorter



Feedlines consists of below components.

6.1.1 Infeed conveyors.

Shipments are loaded on this conveyor in a singulated manner after that it will pass through X- ray machine.

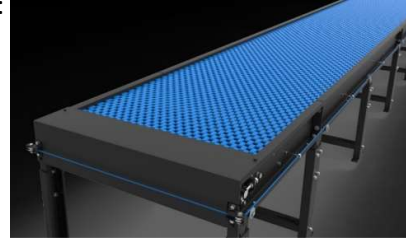


6.1.2 Plastic Modular Belt Conveyor

Falcon's Belt Conveyors are modular & robust in design. MS profile is used to build conveyor frame.

Some Salient Features of Falcon's Modular Belt conveyor:

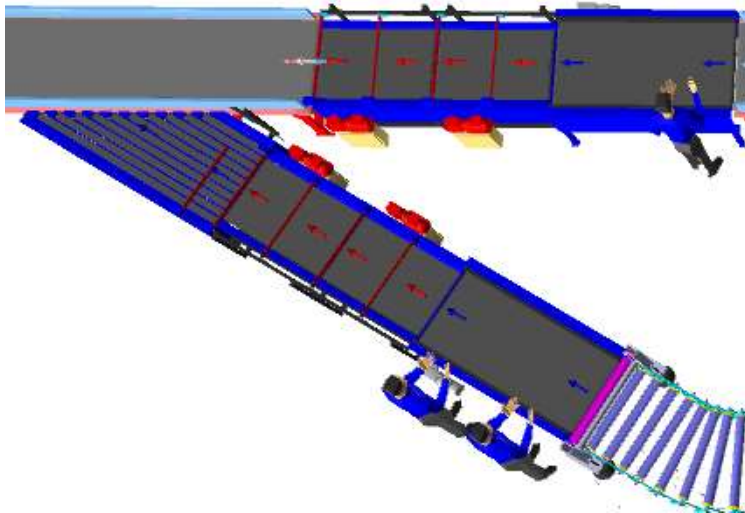
- Low Noise
- Maximum uptime.
- Minimal maintenance
- High safety standards.
- Fastest ROI.



Specification	UOM	Remark
Manufacturer Name	Name	Falcon Autotech
Plastic Modular Belt Material	Type	POM with PBT Pins, Pitch-25.4 mm, Pin Dia- 5mm
Plastic modular Belt Make/Model No.	Make	Movex/XP254 CR
Sprocket Size & No. of Sprockets on Drive End & Tail End	Type	5/146.4 mm
Sprocket material	MOC	POM
Chassis	mm	MS, 3mm Thick, Powder Coated-75 Microns
Wear Strip Material	MOC	UHMW PE1000
Wear Strip Pitch	mm	150 (Strip Width - 40mm)
Return Roller type	Type	DIA- 50 mm/20303
Return Roller Pitch	mm	800
Driven Shaft	Type	MOC-EN9 Solid Shaft hardened to BHN 240, Size-60SQ MM, Zinc Plated -25 microns, Bearings-UCF-210D1
Duplex Sprocket	MOC	MS
Duplex Chain	MOC	MS
Drive Shaft	MOC	EN 24
Conveyor Under guarding	Type	GI SHEET (1MM THICK)
Type of Guides	Type	Steel guides at 50mm/400mm height basis the position of conveyor on both sides from Conveyor Belt Surface.

6.1.3 Shipment Loading Conveyor

It's a approx. 2.5 m long. Powered Belt Conveyor for Shipment Loading. It is running at approx. 800 mm from ground Level.



6.1.4 Detachable flexible conveyor.

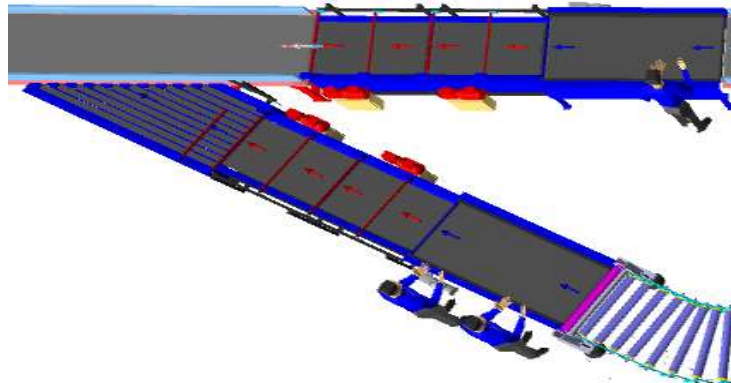
Flexible conveyors are used to load and unload the shipments from truck. It let you change discharge points by rolling and curving conveyor sections to fit your needs. They're commonly used at shipping docks or the discharge end of a conveyor system to connect to truck trailers. A single conveyor can serve multiple dock doors.



S. No	Description of Goods	Details
1	Conveyor Width Between Frame	800mm
2	Conveyor Open Length	10000mm
3	Conveyor Closed Length	5000mm
4	Roller Close Pitch	60mm
5	Roller Open Pitch	120mm
6	Adjustable Height	950 - 1200mm

6.1.5 Orientation Conveyor.

It's a approx. 6 m long. Powered Belt Conveyor for Shipment Orientation running at approx. 800 mm from ground Level.



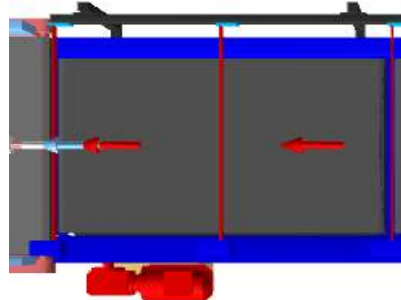
6.1.6 Spacing Conveyor

It's a Special Conveyor, used to create space between parcels



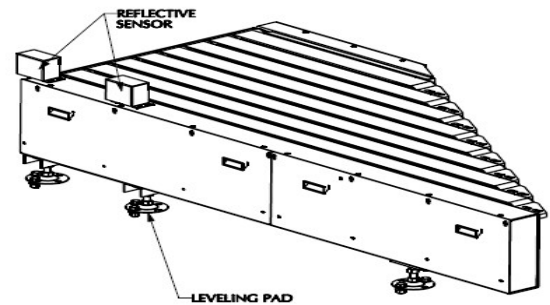
6.1.7 Buffer conveyor

High-speed conveyor, used to create appropriate space between parcels/boxes and synchronize the infeed into the Sorter.



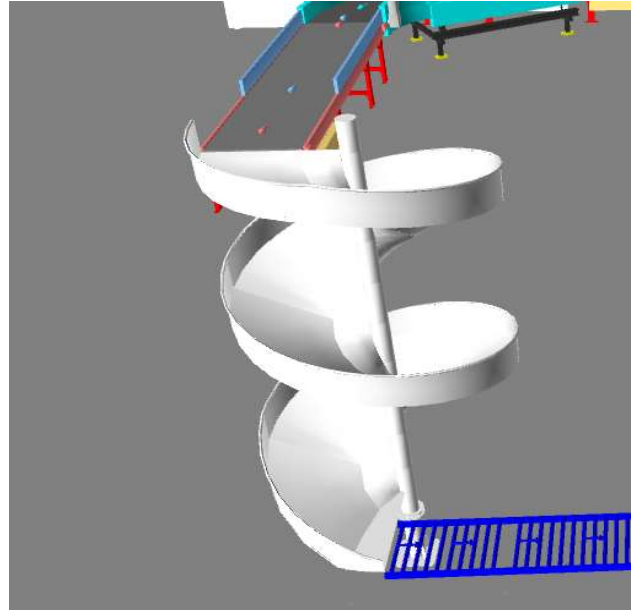
6.1.8 Intelligent merge / Angle merge.

This is 30° triangular Dual Belt high-speed conveyor used for inducting parcel/boxes directly on to the sorter. The Belts are Strip Belts for smooth parcel movement.



6.1.9 Spiral chute.

The Spiral chute is a compact and versatile elevating solution. It is specially designed chute which will take the shipments from G0 to Basement.



6.2 Barcode Scanning

6.2.1 Automatic Barcode Scanning

System is Integrated with Automated Line Scan Camera Technology (Sick ICR) for scanning 1D & 2D Barcodes.



Specification	UOM	Remark
Manufacturer Name	Name	SICK- ICR 890
Barcode Type	Type	1D & 2D
Module Size	Mils	X or higher
Shipment Size	Mm	Min: 150 x 150 x 5 Max: 900 x500 x 500
Number of Codes	Nos	1
Location	Side	Top
Orientation	Type	Top side
Color of bars	Color	Black
VMS	Nos	1
Under Foil	yes/no	No

6.2.2 Handheld barcode Scanner

System is equipped with handheld barcode scanners capable for scanning 1D barcodes for Semi Auto mode.



6.3 Sorter: Swivel Wheel Divert

Swivel Wheel Diverter (SWEDI) is a non-impact sortation module for linear sortation lines, ideal for medium volume sort centres. SWEDI's modular design allows it to save space while also allowing for future expansion of sorting point



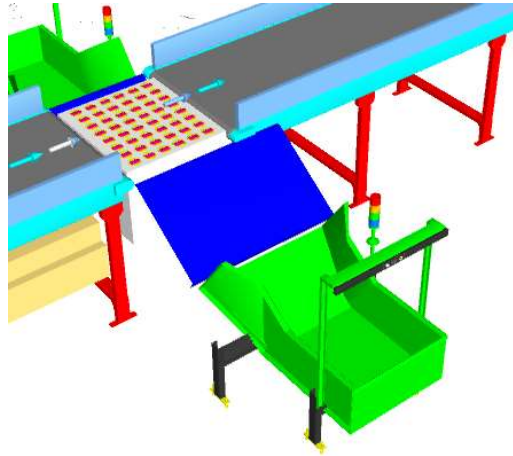
SWIVEL WHEEL DIVERTER (SWEDI) (SMART WHEEL DRIVE)

Technical specifications of SWEDI Unit

Items	Make
SWEDI Unit's Dimensions	815 mm (L) x 915 mm (W)
Load Capacity	35 KG
Designed Throughput	4000 PPH
Conveying Drive Unit	48V, 50 Watt DC powered smart roller
Diverting Drive Unit	Servo Motor; 220V, 50HZ, 750W, Single Phase
Conveying Speed	Up to 120 m/min
Diverting Angle	90 Degrees at double sides
Frame Material	MS Powder Coated Steel Frame
Stand Support	MS Powder Coated Steel Frame
PLC	Siemens
Servo Motor	Delta
Noise Level	<75 Db when measured from 1 m in a silent warehouse.

6.4 Sliding Chutes

Parcels are collected in the in trolley with the help of sliding chute. The chutes are equipped with Chute full sensors and tower lamps.



6.5 Steel works.

Platform 3 SQM (Approximately).



7. Key Component Makes

S. No.	Description	Makes
1	PLC	Siemen's
2	HMI	Siemen's
3	Control Cabinet & Accessories	BCH / Rittal
4	Drives	Allen Bradely/ Lenze/ Siemen's
5	Drive Motor and Gearbox	SEW/ Bonfigloili
6	Sensors	Sick, P&F
7	Bearings	SKF, NBC, NTN
8	PVC Belt	Forbo, Derko,
9	Rollers Make	Pulse/ Falcon
10	Automatic Barcode Scanners	SICK/ Cognex
11	Handheld Scanners	Motorola/ Honeywell

8. Proposed System Technical Details.

Mechanics			
Pos.	Qty.	Description	Value
FM1	1	Conveyor Automation 02 x Angle merge. 13 x Powered Belt Conveyor; L-85.5m 02 x Scanning Conveyor; L-4m 04 x Spacing Conveyor 03 x Buffer Conveyor 03 x Receiving Conveyor 01 x 60-degree merge 01 x Detachable flexible conveyor L- 10m 01 x Modular Conveyor; L- 5.8 m; Width 1200 mm 01 x Gravity Roller Conveyor; L-2m	Width-1000 mm
FM2	1	Falcon Sorter-Swivel Wheel Diverter System Width System Length 1 x Automatic Barcode Scanning System. 07 X Swivel Wheel Divert Units. 06 X Collection chute. 01 X Spiral chute 01 X Rejection. 01 X Sort fail	915 mm 815 mm
Controls			
FC1	1	Controls, IT and Wiring Software Package Control Panels Sorter Control Electrical & Control Hardware Switches Drive panels Operator Panels Internal & Field Wiring	Included

9. Price Schedule

Below chapter consists of Price of Falcon's Mechanics and Controls.

S. No.	Description	Price (USD)
1	Feedline and Sorter	\$ 287,869
2	Conveyors	\$ 168,159
3	Chutes	\$ 5,901
4	Falcon WCS and Integration	\$ 8,684
5	Tool Kit	\$ 1,155
6	Project Management and Engineering Charges	\$ 12,385
7	Packaging & forwarding	\$ 23,178
8	Operator Platform	\$ 470
Grand Total (USD) Excluding Taxes.		\$ 507,802

Note- Above price includes reversible feature in some conveyor modules; Details will be shared in DAP Stage. IT does not includes scanning the reversible direction.

Optional Price for Volume Measurement System is USD 26,600.00

Total Price is to be understood-

- Including Packing & Forwarding Charges
- Excluding Installation & Commissioning Charges
- Freight- Excluded
- Taxes Extra as applicable
- Unloading in Client's Scope
- Laydown Area to be provided by SAB Express.

10.Payment Terms.

The Supplier Quotation is based on the following payment schedule:

Payment %age	Project Stage
50 %	Advance.
50 %	After inspection and clearance before dispatch.

11. Warranty Period

Falcons offered System comes with a standard warranty of 1 year and post completion of warranty, customer can opt for Extended warranty.

12. Delivery Schedule

22 to 24 weeks Ex-Works, Greater Noida from the date of PO, full payment, and DAP.

13. Exclusions

The scope of supply includes all parts which are defined in the Supplier's quotation. All other parts which are not defined in the Supplier's quotation do not belong to the Supplier's scope of supply and are excluded. The following parts are also excluded:

- **Server PC & Server Rack**
- **Excluding installation and Commissioning**
- **Construction Power**
- **Steel Works- If not specified**
- **Collection Bin**
- Building infrastructure; building structure, doors, fire exits, levelling devices, building extinguisher and fire alarm system, building heating and lighting system.
- Electrical power supply and wiring to the main control cabinets
- UPS for Controls and Drives
- Electrical Power for Installation
- Network Cabling up to the Main Server Rack
- Intermediate wiring to parts which are to be supplied by the Purchaser/others
- Emergency/Uninterruptable power supply
- Fire-alarm and fire protection devices
- Traffic and route markings
- Laydown Area
- Ram protection devices
- Cat walks, bridges, maintenance aisles and platforms
- Assembly tools like forklifts and hoisting machines
- All kind of network incl. Local Area Network (LAN/WLAN), exceeding the scope described in Scope of Supply
- Any Kind of Civil work
- Any adjustment of the Supplier's scope of supply to local rules and regulations
- X-Ray machines
- Roller cages/ Pallets
- **Simulation and 3D animation of the sorter system**
- Interface with other equipment not specified in this offer.
- Provision of facilities for the control room (furniture, air conditioning, heating, etc.).
- The supply and installation of fencing around the different corridors.
- Any item specifically indicated as not forming part of the subject matter of the Seller's supply in the offer documentation.